

# Think — Project 1: mixed-up properties

A Cambridge project: which of the four operations obey which laws, tested rather than assumed.

LESSON 33

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 PROJECT 1

LESSON CLOCK

8'

12'

14'

6'

|                         |        |
|-------------------------|--------|
| The three laws          | 8 min  |
| Testing them            | 12 min |
| Counter-examples        | 14 min |
| Presenting the findings | 6 min  |

LEARNING OBJECTIVES

- State the commutative, associative and distributive laws.
- Test each law on each operation with examples.
- Produce a counter-example where a law fails.
- Present the findings as a table with a conclusion.

TEXTBOOK

|           |                        |
|-----------|------------------------|
| National  | pp. 91–93              |
| Cambridge | Stage 7, project pages |
| Workbook  | Project sheet 1        |

01

## Explanation

### The three laws

$a + b = b + a$  $(a + b) + c = a + (b + c)$  $a(b + c) = ab + ac$

| LAW          | WHAT IT SAYS                | IN WORDS                             |
|--------------|-----------------------------|--------------------------------------|
| commutative  | $a + b = b + a$             | the order does not matter            |
| associative  | $(a + b) + c = a + (b + c)$ | the grouping does not matter         |
| distributive | $a(b + c) = ab + ac$        | multiplication spreads over addition |

A LAW THAT HOLDS FOR ONE OPERATION MAY FAIL FOR ANOTHER

That is the whole point of the project: not to be told which laws hold, but to find out by testing.

### Testing them

| OPERATION      | COMMUTATIVE? | ASSOCIATIVE? | TEST USED                |
|----------------|--------------|--------------|--------------------------|
| addition       | yes          | yes          | $3 + 5 = 5 + 3$          |
| multiplication | yes          | yes          | $3 \cdot 5 = 5 \cdot 3$  |
| subtraction    | no           | no           | $3 - 5 \neq 5 - 3$       |
| division       | no           | no           | $6 \div 2 \neq 2 \div 6$ |

ONE EXAMPLE PROVES NOTHING; ONE COUNTER-EXAMPLE PROVES EVERYTHING

$2 + 2 = 2 \cdot 2$  does not make addition and multiplication the same. But  $3 - 5 \neq 5 - 3$  settles for ever that subtraction is not commutative.

Counter-examples

| CLAIM                                    | COUNTER - EXAMPLE                                | VERDICT |
|------------------------------------------|--------------------------------------------------|---------|
| subtraction is commutative               | $3 - 5 = -2, 5 - 3 = 2$                          | false   |
| division is associative                  | $(12 \div 6) \div 2 = 1, 12 \div (6 \div 2) = 4$ | false   |
| subtraction is associative               | $(9 - 4) - 2 = 3, 9 - (4 - 2) = 7$               | false   |
| multiplication distributes over addition | no counter-example exists                        | true    |
| division distributes from the right      | $(6 + 4) \div 2 = 5 = 3 + 2$                     | true    |
| division distributes from the left       | $12 \div (6 + 2) = 1.5, \text{ but } 2 + 6 = 8$  | false   |

A COUNTER-EXAMPLE MUST BE A SINGLE, COMPLETE CASE

“Sometimes it does not work” is not a counter-example. Two numbers, both calculations and the two different answers — that is what settles a claim.

Presenting the findings

| PART OF THE REPORT | WHAT IT CONTAINS                                |
|--------------------|-------------------------------------------------|
| the question       | which laws hold for which operations            |
| the method         | the numbers you tested and why you chose them   |
| the table          | four operations against three laws, with ✓ or ✗ |
| the evidence       | one counter-example for every ✗                 |
| the conclusion     | two sentences on what the pattern is            |

THE PATTERN WORTH NOTICING

Addition and multiplication obey the laws; subtraction and division do not. The two that fail are exactly the two that undo the other two — and that is not a coincidence.

02

## Worked examples

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EXAMPLE 1 Is subtraction commutative? Justify your answer.

1

Test 3 and 5.

2

$3 - 5 = -2$

3

$5 - 3 = 2$

4

The two differ, so no — and one counter-example is enough.

ANSWER

No

**EXAMPLE 2** Is division associative?

① Test 12, 6, 2.

②  $(12 \div 6) \div 2 = 1$

③  $12 \div (6 \div 2) = 4$

④ Different, so no.

ANSWER

No

**EXAMPLE 3** Show that  $a(b + c) = ab + ac$  with  $a = 4$ ,  $b = 5$ ,  $c = 3$ .

① Left:  $4(5 + 3) = 4 \cdot 8 = 32$ .

② Right:  $4 \cdot 5 + 4 \cdot 3 = 20 + 12 = 32$ .

③ They agree — as the distributive law says they must.  
One case, not a proof.

ANSWER

Both give 32

**03 Interactive model**

Ask each group to try to break one law and to bring the counter-example to the board; the groups that fail have found the true laws, which is the more interesting result.

**Which laws hold?**

Addition is commutative:

Subtraction is commutative:

Multiplication is commutative:

Division is commutative:

Addition is associative:

Division is associative:

true

false

$a(b + c) = ab + ac$ :

true

false

One example is enough to prove a law:

true

false

## 04 Terminology

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Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH         | O'ZBEKCHA                  | РУССКИЙ           |
|-----------------|----------------------------|-------------------|
| Commutative     | O'rin almashtirish xossasi | Переместительный  |
| Associative     | Guruhlash xossasi          | Сочетательный     |
| Distributive    | Taqsimot xossasi           | Распределительный |
| Law             | Qonun                      | Закон             |
| Counter-example | Qarshi misol               | Контрпример       |
| To test         | Sinab ko'rmoq              | Проверить         |
| Conclusion      | Xulosa                     | Вывод             |
| Project         | Loyiha                     | Проект            |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Commutative means:

the order does not matter

the grouping does not matter

brackets expand

nothing changes

Associative means:

the order does not matter

the grouping does not matter

signs cancel

the answer is zero

Subtraction is:

commutative

not commutative

associative

distributive



A claim is disproved by:

one example

one counter-example

many examples

a table

The two operations obeying all the laws are:

addition and subtraction

addition and multiplication

multiplication and division

subtraction and division

$(6 + 4) \div 2$  equals:

5

8

3

1.5

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is addition commutative?

ANSWER Yes

2. Is subtraction commutative?

ANSWER No

3. Is multiplication commutative?

ANSWER Yes

4. Is division commutative?

ANSWER No

5. Is addition associative?

ANSWER Yes

6. Is division associative?

ANSWER No

7. State the distributive law

ANSWER  $a(b + c) = ab + ac$

Medium · the standard question

7 PROBLEMS

1. A counter-example to “subtraction is commutative”

ANSWER  $3 - 5 \neq 5 - 3$

2. A counter-example to “division is associative”

ANSWER  $(12 \div 6) \div 2 \neq 12 \div (6 \div 2)$

3. A counter-example to “subtraction is associative”

ANSWER  $(9 - 4) - 2 \neq 9 - (4 - 2)$

4. Check  $a(b + c) = ab + ac$  at 4, 5, 3

ANSWER Both 32

5.  $(6 + 4) \div 2$  and  $6 \div 2 + 4 \div 2$

ANSWER Both 5

6.  $12 \div (6 + 2)$  and  $12 \div 6 + 12 \div 2$

ANSWER 1.5 and 8 — different

7. How many counter-examples disprove a claim?

ANSWER One

### Hard · stretch and reason

7 PROBLEMS

1. Does division distribute over addition from the right?

ANSWER Yes —  $(a + b) \div c = a \div c + b \div c$

2. From the left?

ANSWER No —  $c \div (a + b) \neq c \div a + c \div b$

3. Is  $2 + 2 = 2 \cdot 2$  evidence that the operations agree?

ANSWER No — one case proves nothing

4. Find another pair with  $a + b = ab$

ANSWER 0 and 0

5. Why do subtraction and division fail the laws?

ANSWER They are the inverses of the operations that obey them

6. Is  $a - (b - c) = (a - b) + c$  true?

ANSWER Yes — check with numbers

7. Which of the three laws holds for all four operations?

ANSWER None of them — subtraction and division fail every one

## 07 Homework

### Homework — the project

One page: the table, one counter-example for every cross, and two sentences of conclusion.

1. Draw the table of four operations against the commutative and associative laws.
2. Fill it in with ✓ and ✗, testing each cell with numbers of your own.
3. Write one counter-example beside every ✗.
4. Test the distributive law with three numbers of your choice.

5. Write two sentences saying which operations obey the laws, and what the two exceptions have in common.